

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

WMT Lab Test Submission

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Batch No	-

GitHub Repository Links: https://github.com/dilshan-code/WMT_LabTest_Project.git

Screenshot of the locally running existing project Add Item page

The screenshot shows the 'Add Item' page of the 'Item Manager' application. The page has a dark blue header with the 'Item Manager' logo, a dropdown menu for 'ACTIVE SHEET: All Fields (Master Mode)', and buttons for 'Home' and 'Add Item'. The main content area is titled 'Add Item' and includes a subtitle: 'Complete the fields below. Versioned items 1-22 are logically organized.' Below this is a tabbed interface with four tabs: 'Core Details' (selected), 'Specs & Inventory', 'Physical & Logistics', and 'Pricing & Warranty'. The 'Core Item Details' section contains the following fields: 'ITEM NAME' (Smart Fitness Watch X9), 'CATEGORY' (Electronics), 'PRICE (RS.)' (24999), 'MANUFACTURE DATE' (08/15/2025), 'IMAGE URL' (https://unsplash.com/photos/black-g208-smart-watch-QhF3YGsDrYk), and 'DESCRIPTION' (Premium smart fitness watch with heart rate monitoring, sleep tracking, GPS, Bluetooth calling, and 7-day battery life. Features a water-resistant design, AMOLED display, and compatibility with Android and iOS devices. Ideal for workouts, daily activity tracking, and productivity on the go.). A 'Next Section' button is located at the bottom right of the form.

Screenshot of the locally running existing project Home page

The screenshot shows the 'Item Details Dashboard' page of the 'Item Manager' application. The page has a dark blue header with the 'Item Manager' logo, a dropdown menu for 'ACTIVE SHEET: All Fields (Master Mode)', and buttons for 'Home' and 'Add Item'. The main content area is titled 'Item Details Dashboard' and includes a subtitle: 'View, manage, filter, and search inventory items with integrated Manufacture Date & version details.' Below this is a search bar with the placeholder text 'Search by name, brand, description...'. To the right of the search bar are filters for 'Category: All' and 'Status: All Statuses', along with a '+ Add Item' button. Below the search bar is a message: 'No items match your active filters or search query.' and a 'Clear All Filters' button.

Screenshot of the locally running updated project Add Item page

The screenshot shows the 'Update Item' form in the 'Item Manager' application. The form is titled 'Update Item' and includes a subtitle 'Complete the fields below. Versioned items 1-22 are logically organized.' Below the title are four tabs: 'Core Details', 'Specs & Inventory', 'Physical & Logistics', and 'Pricing & Warranty'. The 'Core Details' tab is active, showing fields for 'ITEM NAME *', 'CATEGORY *', 'PRICE (RS.) *', 'MANUFACTURE DATE *', 'IMAGE URL', and 'DESCRIPTION *'. The 'ITEM NAME' field contains 'Smart Fitness Watch X950', 'CATEGORY' contains 'Electronics', 'PRICE (RS.)' contains '24999.45', 'MANUFACTURE DATE' contains '08/15/2025', 'IMAGE URL' contains 'https://unsplash.com/photos/black-g208-smart-watch-QhF3YGsDrYk', and 'DESCRIPTION' contains 'Premium smart fitness watch with heart rate monitoring, sleep tracking, GPS, Bluetooth calling, and 7-day battery life. Features a water-resistant design, AMOLED display, and compatibility with Android and iOS devices. Ideal for workouts, daily activity tracking, and productivity on the go.' A 'Next Section ->' button is at the bottom right.

Screenshot of the locally running updated project Home page

The screenshot shows the 'Home' page of the 'Item Manager' application. The page features a search bar with the placeholder text 'Search by name, brand, description...', a 'Category' dropdown menu set to 'All', a 'Status' dropdown menu set to 'All Statuses', and a '+ Add Item' button. Below the search bar is a product card for the 'Smart Fitness Watch X950'. The card includes a '10% OFF' badge, a 'Free Shipping' badge, and a '10% OFF' badge. The product is categorized as 'ELECTRONICS' and priced at 'Rs. 24999.45'. The description reads: 'Premium smart fitness watch with heart rate monitoring, sleep tracking, GPS, Bluetooth calling, and 7-day battery life. Features a water-resistant design, AMOLED display, and compatibility with Android and iOS devices. Ideal for workouts, daily activity tracking, and productivity on the go.' The card also shows a 'Manufactured: Aug 15, 2025' date and a 'Show Technical Details' dropdown menu. At the bottom of the card are 'Edit' and 'Delete' buttons.

Frontend Live URL: <https://wmt-labtest-project01.netlify.app>

Backend Live URL: <https://wmtlabtestproject-production-7b99.up.railway.app>

Brief deployment notes including the platforms used

Deployment Notes

1. Frontend (React / Vite)

- **Hosting Platform:** Netlify
- **Source Folder:** /frontend
- **Build Settings:**
 - Base Directory: frontend
 - Build Command: npm run build
 - Publish Directory: frontend/dist
- **Environment Variables:**
 - VITE_API_URL = <https://wmtlabtestproject-production-7b99.up.railway.app/api>

2. Backend (Node.js / Express / Mongoose)

- **Hosting Platform:** Railway
- **Source Folder:** /backend (Configured as **Root Directory** in Railway Build Settings)
- **Build Command:** Auto-detected by Railway Nixpacks (uses Node engine setup in /backend/package.json).
- **Environment Variables:**
 - MONGO_URI = mongodb+srv://Sadinsa:dilSadi2004%40@cluster0.hbla3w9.mongoddb.net/item_manager?retryWrites=true&w=majority
- **Networking:** Port forwarding target configured to **8080** (matching the container's dynamically injected listening port).

3. Database (MongoDB Atlas)

- Hosted fully in the cloud via **MongoDB Atlas**.
- Configured with secure authentication and unrestricted network access permissions for connection requests incoming from the Netlify build system and Railway container.